

PART 1: STATE MACHINE DESIGN

- “What mode is the robot currently in?”
- At any moment, the robot should be in EXACTLY ONE state.
- eg:
 - Following track
 - Avoiding obstacle
 - Parking
- Architecture: PERCEPTION -> EVENTS -> STATE MACHINE -> COMMANDS

Event Table:

EVENT	DESCRIPTION
START_BUTTON_PRESSED	Competition officially starts
PILLAR_RED_DETECTED	Red obstacle found
PILLAR_GREEN_DETECTED	Green obstacle found
OBSTACLE_CLEARED	Obstacle passed successfully
LAP_MARKER_DETECTED	Start section revisited
THREE_LAPS_COMPLETE	Mission complete
FINISH_ZONE_DETECTED	Begin stopping
SENSOR_FAILURE	Enter ERROR

States:

INIT
WAIT_FOR_START
FOLLOW_TRACK (+ obstacle checking)
AVOID_OBSTACLE
LAP_CHECK
FINAL_APPROACH
PARK
STOP
ERROR

State Table:

STATE	GOAL	SENSORS	OUTPUT	ENTRY CONDITION	EXIT CONDITION	NEXT STATE (TRANSITION)
INIT	Initialize all systems safely	Camera, IMU, TOF, serial status	System checks, idle motor commands	Robot powered on	All systems initialized successfully	WAIT_FOR_START
WAIT_FOR_START	Wait legally for competition start	Start button	Motors disabled	INIT completed	Start button pressed after judge signal(“Go”)	FOLLOW_TRACK
FOLLOW_TRACK	Navigate the track while staying centered and moving in correct direction	Camera, TOF, IMU, Wall detection	Steering, speed commands	Start pressed //OR obstacle completed	Pillar detected //OR lap marker detected //OR finish sequence triggered //OR critical failure	AVOID_OBSTACLE //OR ERROR //OR PARK
AVOID_OBSTACLE	Pass obstacle on correct side based on pillar color	Camera, TOF	Modified steering trajectory	Red or green pillar detected	Obstacle successfully passed	FOLLOW_TRACK
LAP_CHECK	Verify and count completed laps	Camera / section detection	Increment lap counter	Lap marker or start section detected	Lap count updated	FOLLOW_TRACK //OR FINAL_APPROACH
FINAL_APPROACH	Return safely to finish/start section after lap completion	Camera	Reduced speed steering control	lap_count == 3	Finish section detected	PARK //OR STOP

PARK	Stop fully within parking/finish zone	Camera	Fine steering, lower speed	Finish section reached	Vehicle fully stopped within target area	STOP
STOP	Maintain complete stop	watchdog monitoring (maybe)	Zero motor speed	Parking complete //OR finish achieved	N/A	N/A
ERROR	Enter safe fail-state	All diagnostic inputs	Stop Motors	Critical failure detected	Manuel reset only	INIT

INIT

Things happening here:

- initialize camera
- initialize serial
- initialize PID
- initialize variables
- verify sensors connected

WAIT_FOR_START

The rules explicitly require:

- switched on
- waiting state
- start button press

FOLLOW_TRACK

Background Processes Running

- track detection
- wall following
- lane estimation
- obstacle detection
- lap detection

AVOID_OBSTACLE

Logic

- Red pillar -> pass RIGHT
- Green pillar -> pass LEFT

LAP_CHECK

Need:

- cooldown timer
- debounce logic

Otherwise:

- double counting happens

FINAL_APPROACH

Robot should:

- become conservative/safer
- prioritize accuracy over speed

PARK

Parking may:

- be optional
- increase score

STOP

Rules specify:

- robot should clearly stop
- avoid ambiguous movement

Possible Triggers

- Camera disconnected
- Serial timeout
- Invalid sensor data
- Fatal software exception

State Priority Ranking

PRIORITY	STATE	REASON
1	ERROR	Safety and fail-safe override all behaviors
2	AVOID_OBSTACLE	Competition-critical obstacle rules
3	FINAL_APPROACH / PARK	Precise mission completion behavior
4	LAP_CHECK	Important but temporary bookkeeping
5	FOLLOW_TRACK	Default navigation behavior
-	WAIT_FOR_START	Pre-run state only
-	INIT	Startup initialization only
-	STOP	Terminal state

Event Priority Ranking

PRIORITY	EVENT
1	CRITICAL_FAILURE
2	PILLAR_DETECTED

3	FINISH_ZONE_DETECTED
4	LAP_MARKER_DETECTED